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Velocity-controlled acoustic tweezers based on beat-frequency modulated surface acoustic waves for programmable particle manipulation

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A B S T R A C T

Achieving continuous, real-time control over particle velocity, not merely position, represents a fundamental challenge in acoustic tweezers technology. While substantial research has focused on precise dynamic manipulation, most platforms lack the simplicity and versatility for direct velocity command. Here, we introduce a velocity-controlled acoustic tweezers platform based on beat-frequency modulated surface acoustic waves (SAWs). This system translates a single, straightforward input—frequency difference—into a precise and direct particle velocity output. We demonstrate precise velocity control for diverse modes of motion, including uniform motion and accelerated motion, precise angular steering at arbitrary orientations, and the execution of complex programmable trajectories (e.g., Lissajous curves). In contrast to phase modulation, which achieve particle manipulation through discrete positional updates, the beat-frequency modulation approach establishes a direct and continuous command over particle velocity, fundamentally simplifying the paradigm of velocity control. We also induced bubble cluster oscillation via beat-frequency modulation. This work establishes a simple (utilizing frequency difference as the sole control parameter) yet powerful framework for direct velocity manipulation, with promising applications in targeted drug delivery and tissue engineering.

1. Introduction

Precise manipulation of microscale particles (such as germs, cells, and microorganisms) is of critical importance in the fields of biomedical engineering, tissue engineering, and diagnostics [1,2]. Traditional methods, including optical tweezers [3], magnetic tweezers [4], acousto-dielectric tweezers [5] and dielectrophoresis [6], have demonstrated remarkable capabilities in particle and cell manipulation. However, these techniques often suffer from limitations such as high power requirements, the need for labeling, or complex setups, which hinder their widespread application in sensitive biological environments. Among emerging technologies, acoustic tweezers, particularly those based on surface acoustic waves (SAWs), have gained prominence due to their label-free, biocompatible, and contactless characteristic. They offer a versatile platform for micromanipulation [7], and enable a range of functions including particle enrichment [8], sorting [9], patterning, and manipulation [10]. Of these capabilities, precise spatiotemporal control of microscale particle trajectories is significant for biomedical applications such as targeted drug delivery and tissue engineering, a feature that has garnered growing research interest [11].

Current particle trajectory control primarily utilizes acoustic radiation forces to trap particles within potential wells, followed by dynamic modulation of the well positions. Qin et al. [12] developed a movable SAW-based tweezers system that employs a translation stage to create relative motion between the piezoelectric substrate and a fixed, independently packaged microchannel, enabling particle translation as well as in-plane and out-of-plane rotation. However, the propagation of acoustic energy through multilayer structures results in low energy conversion efficiency, while the additional translation stage and custom fixtures increase system complexity. In addition, signal modulation methods in SAW-based acoustic tweezers can also be employed for trajectory manipulation of particles [13,14]. Using multi-frequency modulation method, two-dimensional trajectory control of cells and C. elegans was achieved by Ding et al. [15] through dynamically adjusting the input frequencies of two orthogonally chirped interdigital transducer (IDT) pairs to shift pressure node positions. Building on this, Xu et al. [16] theoretically demonstrated, with validation by finite element analysis, that Fourier-synthesized harmonics enable the independent manipulation of multiple particles along distinct 2D paths. However, these approaches exhibit two inherent limitations: (1) The gradually

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varied interdigital spacing in chirped IDTs complicates acoustic modes, resulting in inefficient electromechanical conversion that degrades manipulation precision and acoustic force intensity; (2) The finite number of usable frequencies restricts operational flexibility and spatial resolution, while simply increasing frequency components would further reduce conversion efficiency.

The phase modulation approach has emerged as an alternative method for precise in-plane displacement control. In this method, orthogonal IDT pairs regulate particle trajectories by tuning the phase difference of excitation signals to reposition Gor'kov potential wells [17–19]. Notably, a joint subarray acoustic tweezers platform integrating outer, middle, and inner subarrays was recently developed by Shen et al. [20]. This system achieves not only in-plane displacement through phase modulation but also out-of-plane translation and three rotational motions via SAW-induced acoustic streaming vortices, representing the first demonstration of six degree of freedom manipulation in a single SAW device. However, the phase modulation approach requires path discretization into stepwise motions, with each step's single-axis component constrained to $<\lambda/4$ displacement. Displacements beyond this range cause particles to relocate to the nearest pressure node rather than the target position. More specifically, both multi-frequency and phase modulation are position-based control methods. Each modulation step can only dictate the subsequent target position, making it inherently difficult to exert precise and continuous control over the particle velocity during the translocation process.

The beat frequency phenomenon, a classic wave interference effect, provides a simple approach for direct velocity manipulation [21]. By precisely controlling the slight frequency difference Δf between two counter-propagating traveling waves or orthogonal standing waves, this method directly sets the velocity of particle translation. The resulting beat-frequency acoustic field imparts a precisely tuned, continuous propulsive force that drives particles at a constant speed proportional to Δf. In contrast to multi-frequency modulation and phase modulation (which provide discrete position control), the beat-frequency modulation approach directly regulates particle velocity, significantly simplifying speed regulation. Recent applications have demonstrated its effectiveness in particle sorting [22,23]. Simon et al. developed a size and density/compressibility-based particle sorting technique using on-off quasi-standing waves with high efficiency (>90 %) and purity (>80 %), while also supporting bidirectional sorting [21]. However, the effects of the acoustic field are limited to one dimension, and thus does not yet enable programmable velocity manipulation in a two-dimensional plane.

In this study, we enable programmable particle trajectories in SAW tweezers by governing velocity with beat frequency modulation. Adjusting the frequency difference between counter-propagating traveling waves enables both millimeter-scale uniform motion and accelerated linear motion of particles. Controlling frequency differences in two orthogonal traveling wave pairs allows precise directional regulation. By implementing programmed frequency difference sequences via MATLAB to control the function generator, we demonstrated programmable manipulation along complex paths in a two-dimensional plane. Additionally, controlled beat-frequency oscillations of bubble clusters were achieved by adjusting the frequency difference between two orthogonal standing waves.

2. Methods and material

2.1. Device design and working mechanism

As illustrated in Fig. 1, four IDTs with uniform finger spacing are symmetrically arranged on a lithium niobate ($LiNbO_3$) substrate, surrounding a centrally bonded polydimethylsiloxane (PDMS) chamber that serves as the particle manipulation workspace. Each IDT is rotated by 90° relative to its neighboring set around the focal region. Notably, the SAW propagation axis of each transducer is oriented 45° relative to the X-axis of the $LiNbO_3$ substrate, an alignment that ensures identical SAW velocities from all transducers and thus mitigates anisotropy-related challenges in particle manipulation design [24]. By modulating the input signals (frequency, amplitude, and phase) to each IDT, this configuration enables precise control over the acoustic pressure distribution within the target region, thereby facilitating accurate microparticle manipulation. The system employs pseudo-standing waves (requiring matched amplitudes for signal pairs, e.g., $S_{x,1}$ and $S_{x,2}$) for particle control.

According to Gor'kov potential theory, the acoustic radiation force exerted by standing SAWs on a compressible spherical particle in fluid can be expressed as [25]:

$$\boldsymbol{F}_{\mathrm{rad}} = -\nabla U \tag{1}$$

where U represents the Gor'kov potential generated by standing SAWs, given by:

$$U = \frac{4\pi}{3} r^3 \left(\frac{\alpha_1}{2\rho_{\mathrm{f}} c_{\mathrm{f}}^2} p^2 - \frac{3}{4}\alpha_2 \rho_{\mathrm{f}} \boldsymbol{v}^2\right) \tag{2}$$

where p, $\boldsymbol{v} = \frac{\nabla p}{(2\pi f \rho_{\mathrm{f}})}$ and f denote the acoustic pressure, background sound speed, and frequency, respectively; $\alpha_1 = 1 - \frac{\rho_{\mathrm{f}} c_{\mathrm{f}}^2}{\rho_{\mathrm{p}} c_{\mathrm{p}}^2}$ and $\alpha_2 = 2(\rho_{\mathrm{p}}/\rho_{\mathrm{f}} - 1)/(2\rho_{\mathrm{p}}/\rho_{\mathrm{f}} + 1)$ are dimensionless constants; c_{f} and c_{p} denote the sound speeds of the fluid and particle, respectively.

To illustrate the principle of one-dimensional beat-frequency modulation, we start with two counter-propagating traveling waves of the same amplitude and a slight frequency difference. Their resultant acoustic pressure field is given by:

$$p(x,t) = A_x \left[\cos\left(-\frac{2\pi x}{\lambda_{x,1}} + 2\pi f_{x,1} t\right) + \cos\left(\frac{2\pi x}{\lambda_{x,2}} + 2\pi f_{x,2} t + \Delta\varphi_{x0}\right)\right] \tag{3}$$

where A_x is the amplitude of the SAW propagating along the x-direction;

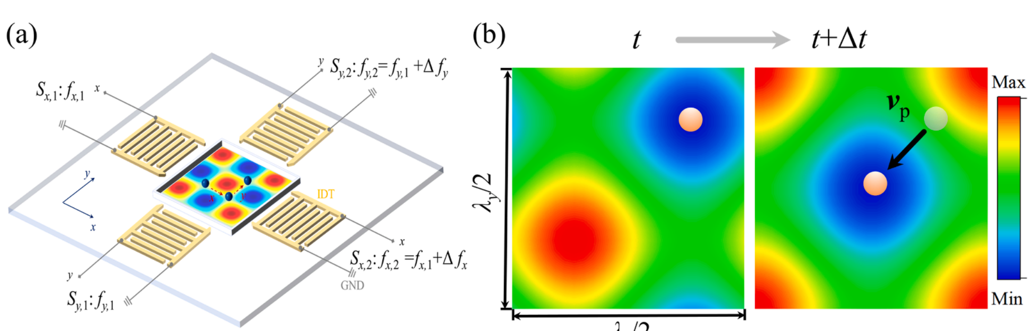

Fig. 1. (a) The schematic diagram of the acoustic manipulation with SAW, which is generated in the x and y directions by four IDTs. (b) Schematic diagram of particle manipulation via the moving pressure nodes. The arrows indicate the direction of motion, and $\boldsymbol{v}_{\mathrm{p}}$ denotes the constant translation velocity of the particles.

$\Delta\varphi_{x0}$ represents the initial phase difference between the two traveling waves; $f_{x,1}$ and $f_{x,2}$ denote the frequencies of $S_{x,1}$ and $S_{x,2}$, satisfying $f_{x,2} = f_{x,1} + \Delta f_x\left(\Delta f_x \ll f_{x,1}\right)$; $\lambda_x = \frac{c_s}{f_x}$ and $c_s = 3589$ m/s are the wavelength and phase velocity of SAW in the $LiNbO_3$ substrate. Using identity $\cos\alpha + \cos\beta = 2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$, Eq. (3) can be simplified to:

$$p(x,t) \cong 2A_x\cos\left(\frac{2\pi x}{\lambda_x} + \frac{\Delta\varphi_x(t)}{2}\right)\cos\left(2\pi\bar{f}_x t + \frac{\Delta\varphi_{x0}}{2}\right) \tag{4}$$

where A_x is the amplitude of the SAW propagating along the x-direction ; $\Delta\varphi_x(t) = \Delta\varphi_{x0} + 2\pi\Delta f_x t$; and $\bar{f}_x = f_{x,1} + \Delta f_x/2$ represents the equivalent frequency. From Eq. (4), the position of the acoustic pressure nodal lines can be derived as:

$$x = \frac{n}{4}\lambda_x - \frac{\Delta\varphi_{x0}}{4\pi}\lambda_x - \frac{1}{2}\Delta f_x\lambda_x t \quad (n = 1, 3, 5\ldots) \tag{5}$$

According to Eq. (5), a small frequency difference in a pair of counter-propagating waves produces a pseudo-standing wave with nodal lines moving at $\boldsymbol{v}_p = \frac{-\Delta f_x\lambda_x}{2}\boldsymbol{e}_x$, consistent with prior experimental observations [21]. Furthermore, the equation shows that the initial phase difference $\Delta\varphi_{x0}$ sets the initial position of the nodal lines. When the frequency difference Δf_x is zero, adjusting $\Delta\varphi_{x0}$ provides static control of the nodal positions, which constitutes the basic principle of phase modulation.

By superimposing two additional counter-propagating traveling waves in the orthogonal direction, with frequencies substantially different from the first pair but maintaining a small inter-wave frequency difference (i.e., $f_{x,1} \cong f_{x,2} \neq f_{y,1} \cong f_{y,2}$), the total acoustic pressure field can be expressed as:

$$p(x,y,t) \cong 2A_x\cos\left(\frac{2\pi x}{\lambda_x} + \frac{\Delta\varphi_x(t)}{2}\right)\cos(2\pi\bar{f}_x t) + 2A_y\cos\left(\frac{2\pi y}{\lambda_y} + \frac{\Delta\varphi_y(t)}{2}\right)\cos\left(2\pi\bar{f}_y t\right) \tag{6}$$

The sound-speed background field is:

$$\boldsymbol{v} = -\frac{2A_x}{\rho_f c_s}\sin\left(\frac{2\pi x}{\lambda_x} + \frac{\Delta\varphi_x(t)}{2}\right)\cos(2\pi\bar{f}_x t)\boldsymbol{e}_x - \frac{2A_y}{\rho_f c_s}\sin\left(\frac{2\pi y}{\lambda_y} + \frac{\Delta\varphi_y(t)}{2}\right)\cos\left(2\pi\bar{f}_y t\right)\boldsymbol{e}_y \tag{7}$$

where $\boldsymbol{e}_x$ and $\boldsymbol{e}_y$ are unit vectors along the x and y directions, respectively. Integrating Eq. (6) and Eq. (7) in Eq. (2) over time gives the averaged Gor'kov potential:

$$\langle U\rangle = C_1\left[\left(2A_x\cos\left(\frac{2\pi x}{\lambda_x} + \frac{\Delta\varphi_x(t)}{2}\right)\right)^2 + \left(2A_y\cos\left(\frac{2\pi y}{\lambda_y} + \frac{\Delta\varphi_y(t)}{2}\right)\right)^2\right] + 4C_2 \tag{8}$$

where, $C_1 = \frac{\pi r^3\left(2\alpha_1 c_s^2 + 3\alpha_2 c_f^2\right)}{\left(6\rho_f c_f^2 c_s^2\right)}$ and $C_2 = -\frac{\pi r^3\alpha_2\left(A_x{}^2 + A_y{}^2\right)}{\left(2\rho_f c_s^2\right)}$. Eq. (8) reveals that two orthogonal pairs of traveling waves with a slight frequency difference (Δf_x and Δf_y) generate pseudo-standing waves whose pressure nodal positions translate at a velocity $\boldsymbol{v}_p = \frac{-\Delta f_x\lambda_x}{2}\boldsymbol{e}_x + \frac{-\Delta f_y\lambda_y}{2}\boldsymbol{e}_y$. In this field, particles with a positive acoustic contrast factor are attracted to the moving nodes by the time-averaged acoustic radiation force and are thus conveyed at the same velocity $\boldsymbol{v}_p$, as demonstrated in Fig. 1(b). To ensure the formation of a clear nodal array rather than complex interference patterns, the frequencies are set to satisfy $f_{x,1} \cong f_{x,2} \neq f_{y,1} \cong f_{y,2}$, thereby preventing coherent coupling between the orthogonal modes [24].

2.2. Fabrication of PDMS microchannel and IDT

The PDMS microfluidic channel was fabricated via standard soft lithography. A 10:1 mixture of PDMS prepolymer and curing agent (Sylgard 184, Dow Corning, Midland, USA) was homogenized at 500 rpm for 20 min, degassed under vacuum, and cast onto a silicon wafer patterned with photoresist (AZ-5200E, AZ Electronic Materials, Merck KGaA, Germany) using image reversal lithography. The patterned silicon wafer underwent secondary degassing followed by thermal curing at 70°C for 160 min in a drying oven (SUPOR, Shaoxing SUPOR Instruments Co., CN). After complete curing, the PDMS chamber was peeled off from the substrate, and access holes were precisely drilled at predetermined locations on the chamber top surface using a punch. The length, width, and height of the microchannel are about 3 mm, 720 μm, and 100 μm, respectively.

The transducers were fabricated on a 3 mm thick, double polished 128° Y-cut $LiNbO_3$ substrate. Four IDT were patterned via photolithography, followed by electron beam evaporation (ULVAC EI-5Z, Abner, CN) of a 5 nm Cr adhesion layer and a 50 nm Au electrode layer. The IDT design consisted of 20 finger pairs with a uniform width and spacing of 90 μm and a 2 cm aperture. The substrate was then diced into individual dies using an LED UV laser slicing machine. The fabricated IDT exhibited a fundamental resonance frequency of ~9.9 MHz.

2.3. Experimental setup

The SAW tweezer device was fixed on an optical microscope stage (M330BD-HD228S, CN). After manually injecting particle solution into the microchannel using a syringe, particle motion was recorded via a CCD camera connected to a host computer. For acoustic excitation, two controllable AC signals generated by a dual-channel function generator (33622 A, Keysight, USA) were amplified by power amplifiers (2100 L, EI, USA) and connected to one pair of IDTs. The orthogonally arranged IDT pair received similar power configuration. Signal parameters could be adjusted either manually via the generator interface or through programmed MATLAB control for complex sequences.

2.4. Particle and microbubble sample preparation

The polystyrene (PS) microsphere suspension (UniFlu PS, Wuxi Rigor Biotechnology Co., CN; material properties: density $\rho_p = 1065$ kg/m³, speed of sound $c_p = 2066$ m/s, Poisson's ratio 0.355) was diluted to working concentration using deionized water containing 2 % surfactant (Tween 80, Harveybio, CN; material properties: density $\rho_f = 1001$ kg/m³, speed of sound $c_f = 1505$ m/s). The lyophilized ultrasound contrast agent microbubbles (SonoVue, Bracco Imaging S.p.A., IT) were reconstituted by injecting 5 mL of ultrapure water into the vial and then shaking vigorously. The resulting stock suspension was then diluted to the working concentration using ultrapure water.

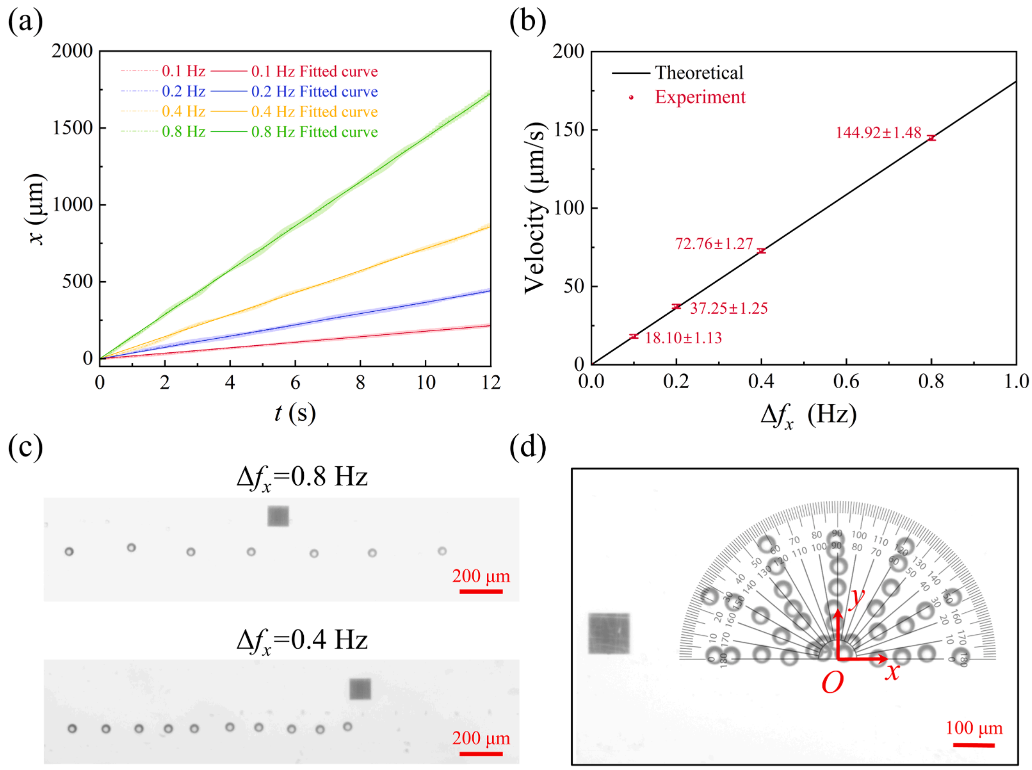

Fig. 2. Particle velocity and direction control based on beat-frequency modulated method. (a) Measured time-position curves under different frequency differences Δf_x. (b) Experimental and theoretical particle velocities for various frequency difference Δf_x. (c) Superimposed frames from the $\Delta f_x = 0.8$ Hz and 0.4 Hz videos at two-second intervals, respectively. (d) Overlay images from corresponding video showing programmable angular steering (0°–180° in 30° increments).

3. Results and discussion

3.1. Programmable particle translocation with tunable velocity and steering

The experimental parameters were set as follows: $f_{x,1} = 9.91$ MHz, $f_{y,1} = f_{y,2} = 9.94$ MHz input voltage was fixed at 30 V. The y-direction signal was applied to constrain particle displacement along this axis. A frequency of 9.94 MHz (rather than 9.91 MHz) was selected to meet the prerequisite $f_{x,1} \cong f_{x,2} \neq f_{y,1} \cong f_{y,2}$ for uniform nodal movement. This configuration creates periodic potential wells (spacing $= \lambda/2$) for particle trapping. The capability of beat-frequency modulation in SAW tweezers to regulate particle speed and direction is demonstrated in Fig. 2. Fig. 2(a) shows time-position curves of 40 μm PS particles under different frequency differences Δf_x (0.1 Hz, 0.2 Hz, 0.4 Hz, and 0.8 Hz), with the shaded bands representing the standard deviation. Dashed lines represent experimental averages, while solid lines show linear fits. All curves presented in this work are derived from data collected through at least five independent experimental trials, ensuring the statistical robustness of the results. In addition, all error bars and shaded error bands correspond to one standard deviation. The particles trajectories are extracted from recorded videos (50 fps) using Tracker software's continuous tracking function.

Fig. 2(b) plots the average particle velocities over a 12-second period obtained from the experimental data in Fig. 2(a), together with the corresponding theoretical values calculated using Eq. (5). Quantitative analysis confirms a linear relationship between velocity and frequency difference within a defined range, with the experimental results closely matching the theoretical prediction and errors remaining within 3 %. Representative results are shown in Supplementary Video V1. As evidenced by the above results, the beat-frequency modulation approach enables continuous and stable operation across millimeter-scale ranges. In contrast to conventional phase modulation approaches, which rely on controlling phase intervals and time delays to achieve equivalent velocity regulation, this method eliminates the $\lambda/4$ step-size limitation and enables real-time, direct velocity control. Fig. 2(c) shows superimposed frames from the $\Delta f_x = 0.8$ Hz and 0.4 Hz videos at two-second intervals, respectively. The black square (100 μm edge length) serves as a fiducial marker on the $LiNbO_3$ substrate, facilitating PDMS chamber alignment and providing a stationary reference for dynamic video tracking. Visual inspection of the plot confirms close agreement between the experimental results and theoretical expectations.

Furthermore, the well-established linear relationship between particle velocity and frequency difference allows straightforward determination of the required frequency difference ratio ($\Delta f_x/\Delta f_y$) for target movement direction. For instance, by programming two orthogonal IDTs with specific parameters—setting $\Delta f_x = -0.4$ Hz, $\Delta f_y = -0.231$ Hz (where $\Delta f_x/\Delta f_y = \sqrt{3}$), and ensuring simultaneous signal excitation—the particles can be driven to move along a 30° trajectory toward the target position. Similarly, varying the frequency difference ratio enables steering at different angles. In Supplementary Video V2, we demonstrate programmable steering across angular ranges (0°–180° in 30° increments), with Fig. 2(d) showing the corresponding overlay images.

To perform a comparative analysis between the proposed beat-frequency modulation method and the phase modulation method, both approaches were evaluated under identical straight-line motion experimental conditions. It should be emphasized that in phase modulation, equivalent velocity control of the particle necessitates the coordinated adjustment of two key parameters: the phase step $\Delta\varphi_m$ (the magnitude of phase shift per modulation) and the time interval Δt (the time interval between consecutive modulation actions). Based on Eq. (5), the theoretically equivalent velocity under phase modulation can be expressed as $-\Delta\varphi_m \lambda_x/(4\pi\Delta t)\cdot \boldsymbol{e}_x$. Fig. 3(a) and its magnified view present the experimentally obtained position-time curves under three modulation strategies, all sharing the same theoretical velocity of 72.04

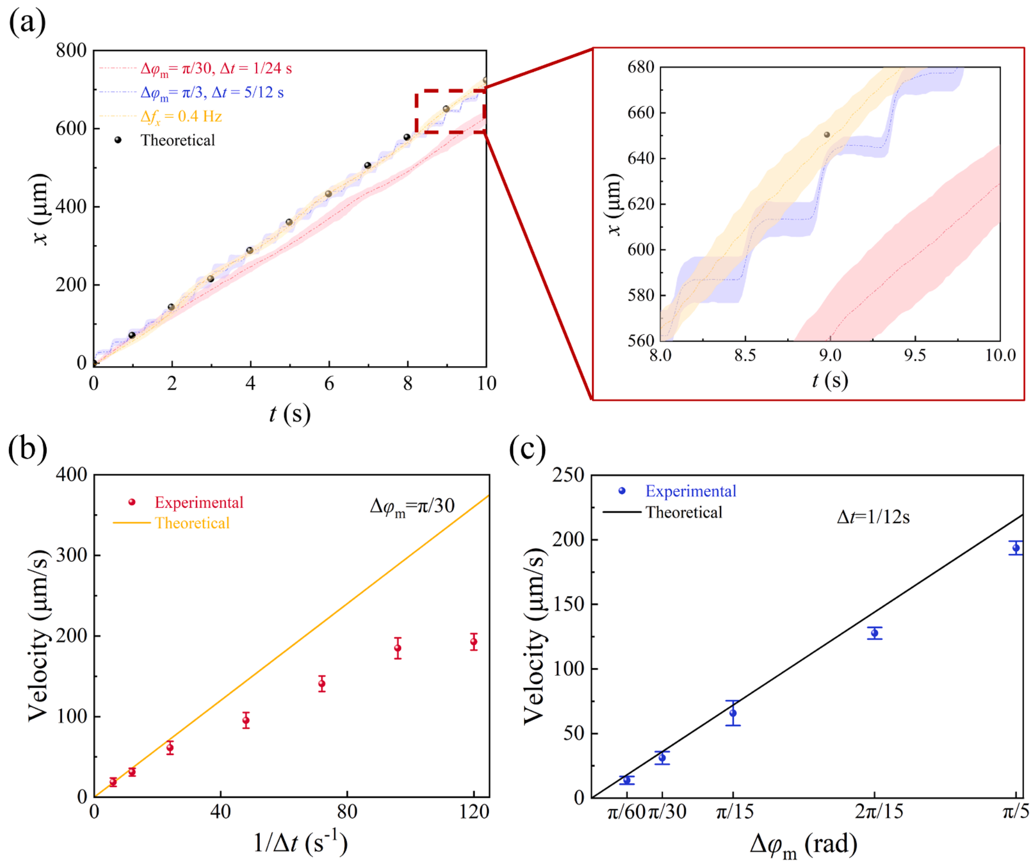

Fig. 3. (a) Position-time curves under different modulation strategies. (b) Experimental and theoretical particle velocities for various $1/\Delta t$. (c) Experimental and theoretical particle velocities for various phase step $\Delta\varphi_m$.

μm/s. The corresponding experimental results obtained using the beat-frequency modulation method ($\Delta f_x = 0.4$ Hz) and the phase modulation method ($\Delta\varphi_m = \pi/3$, $\Delta t = 5/12$ s) are provided in Supplementary Videos V1 and V3, respectively. Experimental results demonstrate that when the modulation phase step is large ($\Delta\varphi_m = \pi/3$), the particle no longer maintains uniform motion but instead exhibits a stepwise ascending pattern. Although the overall velocity trend aligns with the preset value, the discontinuous nature of node position shifts in phase modulation method—which fundamentally operates by discretely altering the positions of acoustic pressure nodes—leads to intermittent velocity during particle motion. Specifically, the particle briefly dwells at each target node after arrival, awaiting the next phase update, consequently creating velocity discontinuity between adjacent nodes.

While reducing the modulation time interval could theoretically smooth the particle's movement between nodes, this approach results in an actual velocity lower than the theoretical prediction, as shown in Fig. 3(a) and (b). When velocity is controlled by varying the time interval Δt under a fixed modulation phase step ($\Delta\varphi_m = \pi/30$), the experimental velocity diverges markedly from the theoretical linear prediction with respect to $1/\Delta t$. Instead, it demonstrates marked nonlinearity, with the deviation from the theoretical value increasing as $1/\Delta t$ rises. This discrepancy stems from the fundamental principle of phase-based method, which operates by repositioning acoustic pressure nodes. Consequently, the resulting velocity error is governed by both the spatial precision of node placement and the temporal inaccuracies inherent to the phase-stepping process.

For comparison, Fig. 3(c) presents the relationship between the particle velocity and the modulation phase step under phase modulation at a fixed time interval ($\Delta t = 1/12$ s). The results indicate that while the velocity exhibits an approximately linear relationship with the modulation phase step $\Delta\varphi_m$ when the modulation time remains constant, a considerable deviation persists between the experimental velocities and theoretical predictions, with error rates surpassing 10.3 %. This error rate is significantly higher than the 3 % observed with the beat-frequency modulation method. In summary, phase modulation exhibits certain limitations in velocity control, primarily manifested as a nonlinear relationship between velocity and $1/\Delta t$, along with a relatively high overall error rate. In contrast, beat-frequency modulation demonstrates superior control performance: it maintains an excellent linear relationship between the phase difference and velocity, and the experimental results agree well with theoretical predictions. Therefore, beat-frequency modulation not only offers advantages in velocity control but also effectively enhances the overall control accuracy of both the position and velocity of particles.

3.2. Programmable particle translocation with uniformly accelerated motion

The beat-frequency modulation method demonstrates superior capability in controlling nodal velocity, enabling the realization of uniformly accelerated linear particle motion through continuous frequency modulation (see Supplementary Video V4), which is more challenging to achieve via phase modulation approaches. In the experiments, the frequency of the input signal was linearly varied over time through parameterized programming. Fig. 4(a) displays the time-position relationships of 40 μm PS particles under varying frequency difference sweep rates ($\Delta\dot{f}_x = 0.1$, 0.15 and 0.2 Hz/s). The experimental results demonstrate successful particle acceleration along linear trajectories, with measured accelerations values of 18.7, 27.3 and 36.6 μm/s^2 correspond well with their theoretical counterparts of 18.1, 27.1, and 36.2 μm/s^2, respectively, with a maximum relative error of less than 3.5 %. Fig. 4(b) shows the overlay image of $\Delta\dot{f}_x = 0.2$ Hz/s case at 1 s

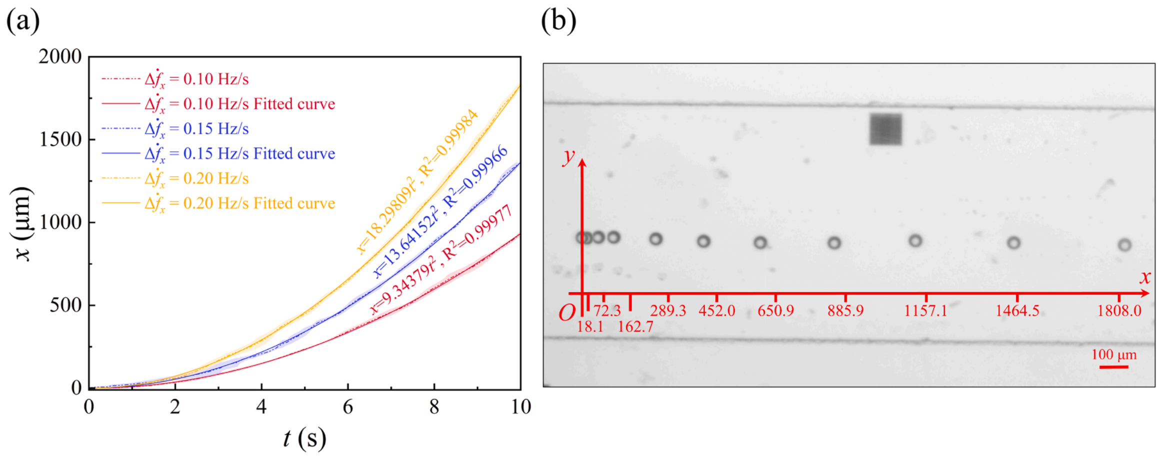

Fig. 4. Uniformly accelerated linear motion of particles. (a) Experimentally measured time-position curves under different frequency difference sweep rates $\Delta\dot{f}_x$. (b) Superimposed frames from corresponding video at 1 s intervals.

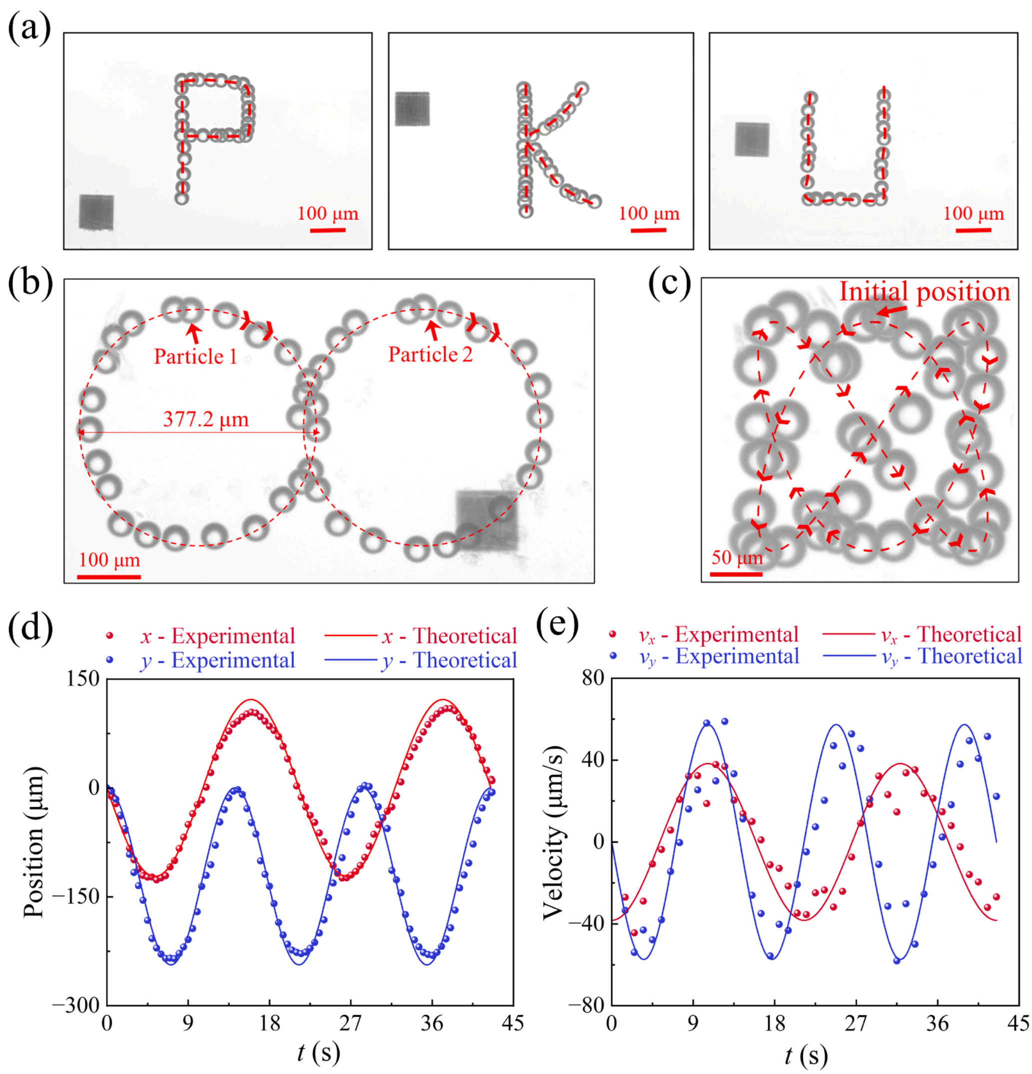

Fig. 5. Complex trajectory manipulation of particles. (a–c) Examples of programmed paths: the letters 'P', 'K', 'U', circles, and Lissajous curves. (d, e) Corresponding time evolution of the (d) position and (e) velocity for a particle following the Lissajous trajectory shown in (c).

intervals, the tick marks on the red horizontal axis correspond to the respective theoretical positions.

3.3. Programmable particle motion along complex trajectories

The aforementioned experiments validate precise control over both the magnitude and direction of particle velocity, establishing the foundation for complex trajectory manipulation. By decomposing target paths into sequential linear displacement vectors, extended and intricate translational trajectories can be programmed. Specifically, the alphanumeric characters 'P', 'K', and 'U' were segmented into elementary linear components. The requisite frequency parameters—including frequency differences and their corresponding durations—were calculated for each vector segment and dynamically implemented through real-time tuning of the IDTs excitation frequencies. This approach successfully guided 40 μm PS particles along all the three character paths (Fig. 5(a) and Supplementary Video V5). Using the character 'U' as an illustrative example, the trajectory was generated through sequential application of the following frequency differences: initial vertical descent was achieved with $\Delta f_x = 0$ Hz, $\Delta f_y = 0.1$ Hz for 16 s; subsequent horizontal translation employed $\Delta f_x = 0.1$ Hz, $\Delta f_y = 0$ Hz for 12 s; and final vertical ascent was accomplished with $\Delta f_x = 0$ Hz, $\Delta f_y = -0.1$ Hz for 16 s to complete the character formation.

The implementation of automated MATLAB modulation facilitated the execution of more complex trajectories. For a circular path with a design radius of 190 μm, the circumference was discretized into 100 linear segments. The particle was guided along each segment n by applying driving signals with frequency differences $\Delta f_x = -A\sin(0.02\pi n)$ and $\Delta f_y = A\cos(0.02\pi n)$, where the amplitude $A = 0.1\pi$ Hz determined the linear velocity. Each segment was maintained for a duration of 0.21 s. Using this approach, two 40 μm PS particles trapped at distinct pressure nodes were simultaneously guided along coordinated circular trajectories (Fig. 5(b) and Supplementary Video V6). The experimentally measured trajectory radius was approximately 188.6 μm, showing close agreement with the theoretical design value of 190 μm.

A Lissajous curve represents a stable, closed trajectory resulting from the superposition of two orthogonal harmonic oscillations with simple integer frequency ratios. In this work, the Lissajous trajectory was discretized into 200 linear segments, each maintained for 0.21 s. The particle was guided along each segment n by configuring the acoustic field drive signals as $\Delta f_x = 0.2\cos(0.02n\pi)$ Hz and $\Delta f_y = 0.3\sin(0.03n\pi)$ Hz. Using this modulation scheme, 40 μm PS particles were successfully transported along a Lissajous trajectory with a frequency ratio of 2:3 and a phase difference of $\pi/2$, as illustrated in Supplementary Video V7, with the corresponding time-lapse composite image presented in Fig. 5(c).

The position-time and velocity-time curves of the particle following a Lissajous trajectory were extracted and compared against the theoretically preset values, as shown in Fig. 5(d) and (e). Fig. 5(d) displays the experimental time-position curves of the particle along the x-y directions, showing good agreement with the theoretical trajectory. This indicates that the system can effectively respond to complex input signals. Fig. 5(e) presents the corresponding time-velocity curves derived from the positional data, whose variation trends and amplitude are consistent with theoretical predictions. We acknowledge that the precision of velocity control for complex trajectories such as Lissajous patterns is somewhat reduced compared to simple linear motion. This is primarily due to their inherent complexity: (1) Lissajous trajectories require continuous changes in both the magnitude and direction of the velocity vector, which is intrinsically more challenging than controlling linear motion; (2) During such dynamic motion, the influences of factors such as acoustic streaming, particle inertia, minor particle imperfections, and imperfections in of the acoustic field are amplified, making the particle more prone to deviation from the ideal path.

Despite these challenges, the beat-frequency modulation method employed in this work offers distinct advantages for realizing complex trajectories—especially those requiring continuous and fine velocity modulation. This method enables smooth motion by implementing gradual and continuous adjustments of the frequency difference. In contrast, conventional phase-based control involves a discontinuous “start-stop” or “accelerate-decelerate” sequence at each step. The associated inertial effects are more pronounced in such schemes, which can introduce greater dynamic instability during complex trajectory manipulation. In summary, although achieving high-precision speed control for complex trajectories remains a significant challenge, the experimental data demonstrate the feasibility of the beat-frequency modulation method, which offers a distinct advantage in realizing smooth and continuous speed control.

3.4. Beat-frequency oscillation

While the aforementioned method controls particle transport via the frequency difference between counter-propagating traveling waves, introducing a frequency difference between two classical orthogonal standing waves allows for dynamic modulation of the acoustic potential wells themselves. Two classical orthogonal standing waves (i.e., $f_{x,1} = f_{x,2} \equiv f_x$, $f_{y,1} = f_{y,2} \equiv f_y$) can generate an acoustic pressure field expressed as:

$$p(x, y, t) = A_x \cos\left(\frac{2\pi x}{\lambda_x}\right)\cos(2\pi f_x t) + A_y \cos\left(\frac{2\pi y}{\lambda_y}\right)\cos\left(2\pi f_y t\right) \tag{9}$$

When a slight frequency difference exists between the two orthogonal standing waves, i.e. $f_y = f_x + \Delta f(\Delta f \ll f_x)$, the time-averaged Gor'kov potential can be expressed as [26]:

$$\langle U \rangle \cong C_1\left[A_x{}^2\cos^2\left(\frac{2\pi x}{\lambda_x}\right) + A_y{}^2\cos^2\left(\frac{2\pi y}{\lambda_y}\right) + C_3 A_x A_y \cos\left(\frac{2\pi x}{\lambda_x}\right)\cos\left(\frac{2\pi y}{\lambda_y}\right)\cos(\Delta\varphi(t))\right] + C_2 \tag{10}$$

where, $C_3 = \frac{4\alpha_1 c_s^2}{(2\alpha_1 c_s^2 + 3\alpha_2 c_f^2)}$ and $\Delta\varphi(t) = 2\pi\Delta f t$. It can be observed from Eq. (10) that introducing a minute frequency difference between orthogonal standing waves enables periodic modulation of the acoustic potential distribution, with a modulation period of $\frac{1}{\Delta f}$.

Here, bubble clusters were selected as control targets owing to their high acoustic contrast factor relative to the surrounding medium and their ultralow density, which promote pronounced dynamic response under acoustic excitation. As established in the literature [27], the phase difference between two orthogonal standing waves of identical frequency critically influences the distribution of the acoustic potential field, as demonstrated in Fig. 6(a)-(c). Introducing a slight frequency difference can be interpreted as continuous temporal modulation of the phase difference, effectively realizing beat-frequency modulation of the acoustic potential field, as derived in Eq. (10). To experimentally validate this mechanism, a frequency difference of 1 Hz was applied, successfully driving bubble clusters into oscillatory motion at the beat frequency (Supplementary Video V8). Experimental observation snapshots are presented in Fig. 6(d)–(f), which not only match the simulated potential well profiles closely but also have a modulation period consistent with the theoretical value. It is noteworthy that strong

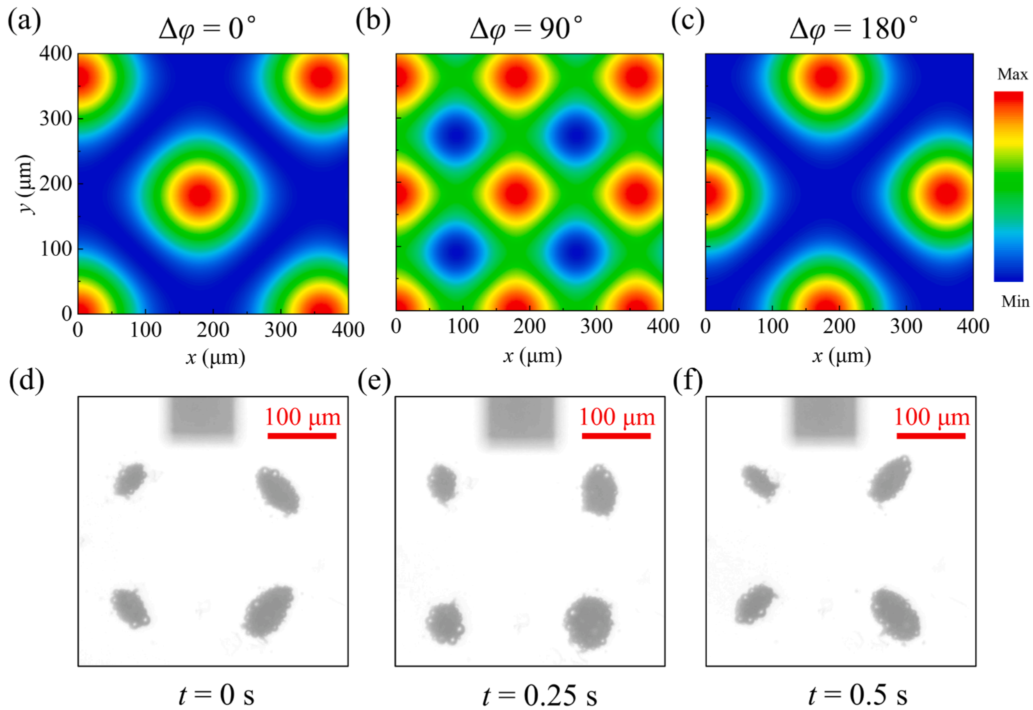

Fig. 6. Beat-frequency oscillation of bubble clusters. (a)-(c) Simulated Gor'kov acoustic potential field distributions under deferent phase differences. (d)-(f) Experimentally observed bubble cluster at selected time point.

secondary acoustic radiation forces among individual bubbles induce tight clustering [28], which causes slight morphological distortion of the clusters relative to the shape of the underlying potential well.

4. Conclusions

This work introduces a SAW tweezers technique based on beat-frequency modulation. By applying a controlled frequency difference between two opposing IDTs, we achieved reliable and stable velocity control of particle motion. This method enables precise regulation of both particle velocity and acceleration through flexible adjustment of the frequency difference. In contrast to conventional phase-modulation strategies that rely on positional control of acoustic pressure nodes, the beat-frequency modulation approach ensures continuous and accurate velocity regulation, thereby achieving high precision in both positional and velocity control. Furthermore, by integrating a custom-developed MATLAB control program, we demonstrated precise directional steering and stable execution of complex programmable trajectories, including Lissajous curves, with excellent control accuracy. Additionally, this work successfully exhibited beat-frequency-induced oscillation of bubble clusters. The proposed modulation strategy offers a simple yet effective method for manipulating microparticle trajectories, significantly expanding the functional capabilities of acoustic tweezers without increasing hardware complexity or cost.

CRediT authorship contribution statement

Duo Xu: Writing – original draft, Resources, Methodology, Investigation, Data curation. **Lin Wu:** Visualization, Software, Data curation. **Yuyang Lin:** Validation, Supervision. **Chunyu Xu:** Writing – review & editing, Software, Project administration, Methodology, Investigation, Formal analysis. **Yongmao Pei:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.eml.2026.102451.

Data availability

Data will be made available on request.

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